NETWORKING

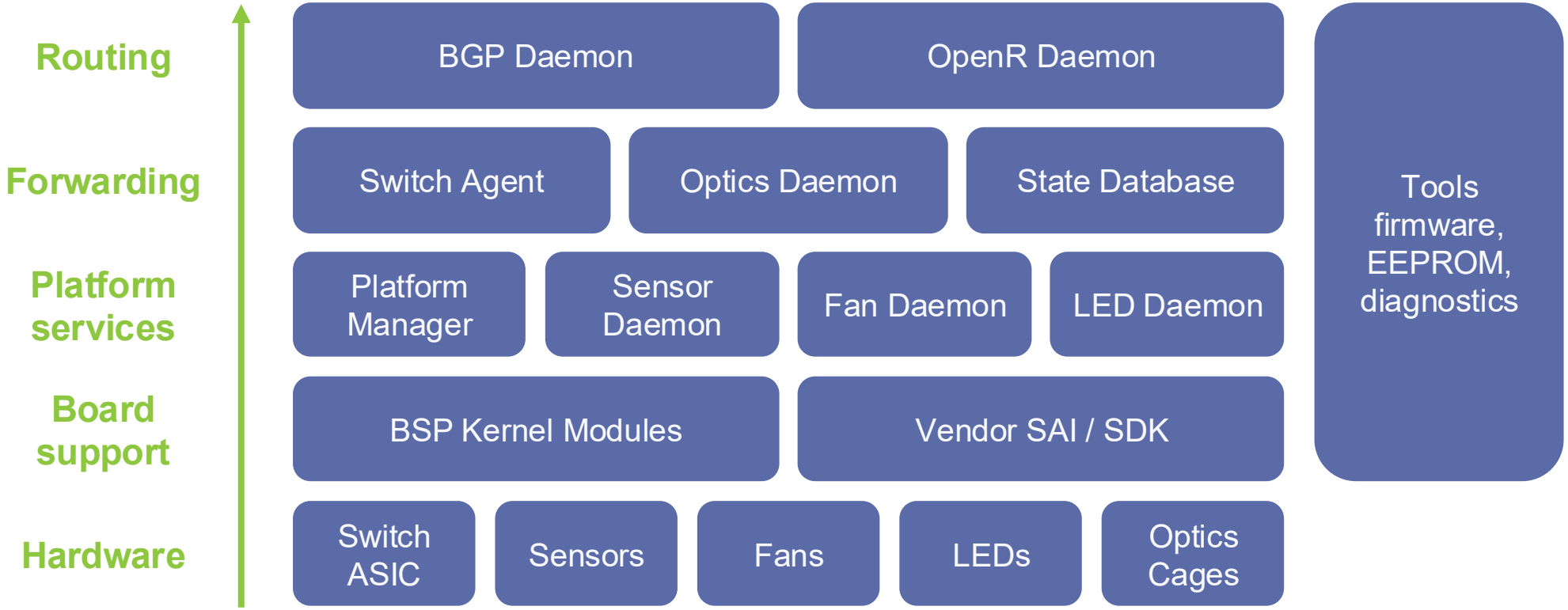


Config-Driven FBOSS: From Board Bring-up to Qualified Platform

Benoit Sigoure *Software Engineer @ Nexthop AI*

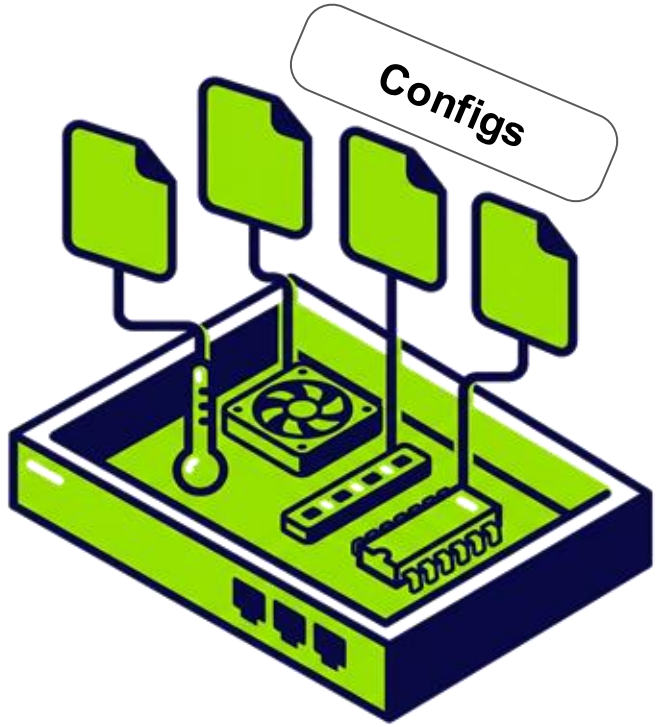
Srikrishna Gopu *Software Engineer @ Meta*

What runs on an FBOSS switch



Config-driven FBOSS services

	Before	Now
Platform services	Service code per board	One JSON file per subsystem
Switch agent	Two dozen files; rebuild to retune	One descriptor, loaded at run time
Optics	Lane maps written by hand	Derived from SFF-8024



Buildable and qualifiable in open source

- Routing in open source — BGP and OpenR
- BSP and kernel modules in open source
- Every service builds from source, one toolchain
- Test artifacts build alongside the services

Source to qualified platform, entirely in open source

Deploying and configuring, vendor-side

FBOSS Distro

- One artifact carries every service and its tests
- Startup order and config ship with the image
- Wipe and re-image to a known state in minutes

Config CLI

- An industry-familiar CLI instead of raw JSON
- Dependent config updated together, with rollback
- Hitless where possible; the CLI knows what needs a coldboot



One artifact, every service

Ordering bugs surface on the first boards · config changed with no request to Meta

Topology-based solution testing

Provision with FBOSS Distro

Snake — One DUT

- **Loop ports back**
- **Drive traffic**
- **Health checks**

Solution — Multi-node

- **Topology + BGP**
- **Traffic + disruption**
- **Health checks**

Qualified - All Suites Passing

Test as a Config: scenarios declared as data, not scripts
The framework drives setup, traffic, disruption and health checks

Qualify a new platform by supplying topology data, not new tests

Open source, Apache 2.0 — works with any traffic generator. Vendor onboarding guide published.

How AI brought up the platform

AI-Driven NPI Bring-Up Workflow

- **Spec to Schema:** AI skill feeds platform HW spec directly to descriptor JSON.
- **5 Minutes Per Iteration:** Fast cycle time for a fully cleaned system.
- **Automated Triage:** Repeatable image load & logging enables AI failure diagnosis.
- **Rapid Engineering:** Network engineers iterate network configs & build UIs rapidly.

Multi-Node Testing (8 × 800G Traffic Hashing)

FBOSS pushing 8 × 800G to peer — 1000 loopbacks with even traffic hashing.




```
[root@crow110 ~]# fboss2 show bgp stats entries
BGP entry statistics:
 Total number of unicast routes: 1000
 Total number of rib paths: 8000
 Total number of adjribs: 8
 Total number of originated routes: 0
```

EXAMPLE AUTOMATED TRIAGE

Observed Scope: Snapshot at ticket creation time.

Frequency: 18 occurrences in 1 batch (bgpd boot-loop incident)

Testbed: crow224 | **Branch:** main

Failure Type: Product bug

Current Spread: Same as creation

Analysis: Automated root-cause isolation complete.

Observability & Debug Stack for FBOSS Qualification

Routing chain validator



Agent crashes & incidents



Live syslog streaming & auto-diagnostics



Results: Wedge 800 and 102T

TAAC testing discovered FBOSS service and product firmware bugs!

	25.6T Wedge 800	102T- TH6P
Bring-up → TaaC validated	Yes, single & multi-DUT	Yes, single DUT
Suites passing	235 tests passing, 50 failing, 200 being onboarded	50 tests passing, 400 tests being onboarded
Done without Meta	No	Yes
Still open	Complete open sourcing the tests	Set up the multi-DUT env

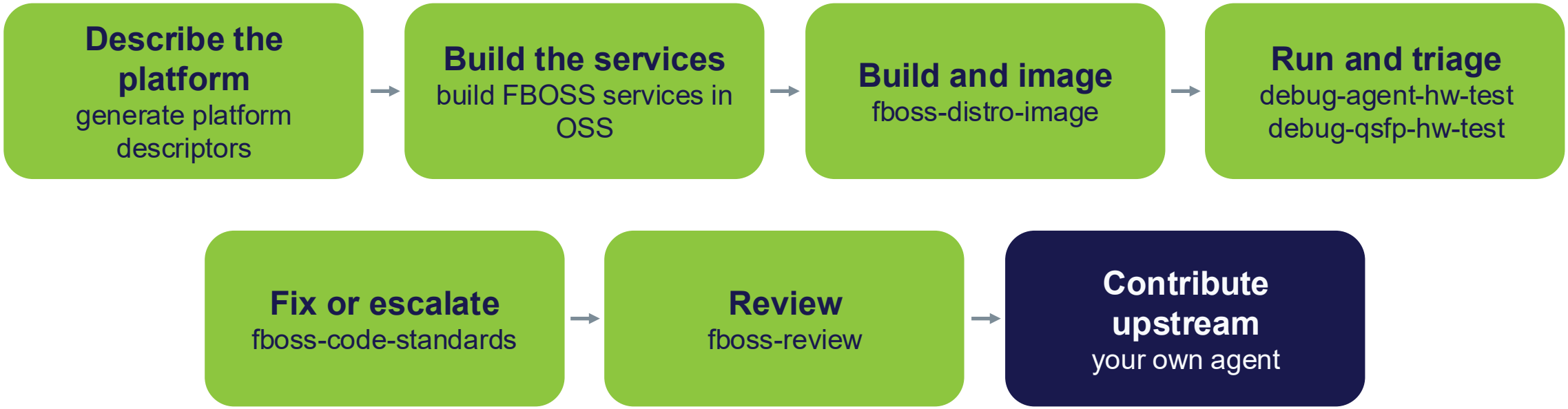


25.6T - Wedge 800



102T - TH6-P

FBOSS AI skills onboarding workflow

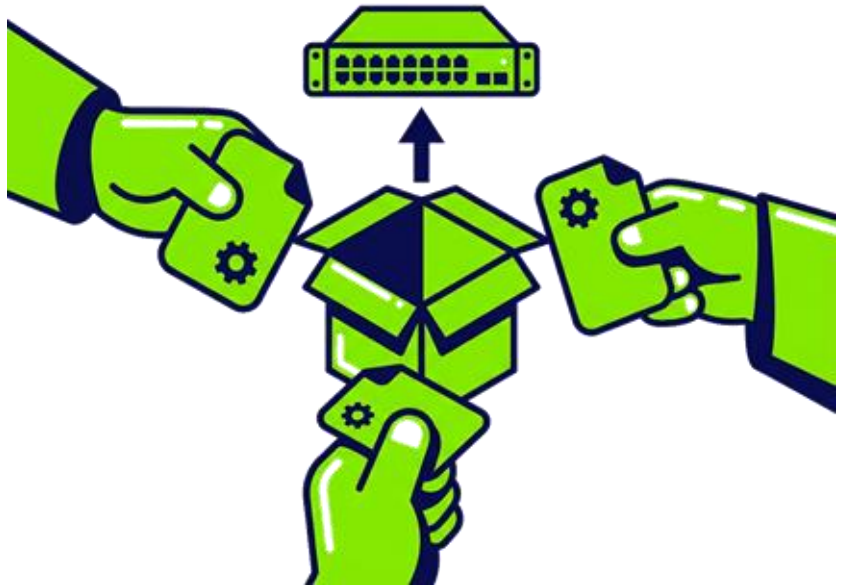


In open source today

Plain text — works with any coding agent. Extend with your own overlay, no fork required.

Call to action

- **Adopt** — Build from source, describe your platform in config, image from day one
- **Automate** — Use the skills with the agent you already use
- **Contribute** — Descriptors, Schemas, Tests, New skills



OCP Contribution:

- Wedge800 Contribution: https://www.youtube.com/watch?v=Ue_309TStyQ
- Enabling networking switch designs through FBOSS - https://www.youtube.com/watch?v=Tzf_mV8QYCg

FBOSS Wiki: https://facebook.github.io/fboss/

Thank You!

